Question 2.a

As a financial consultant, you are given the following information about two companies. [Numbers in grey color are given, numbers in bold are answers].

Fill in the missing figures given that the expected return of the market portfolio is 12% and the risk-free rate is 2%. Assume that there is no corporate tax rate and all cash flows are perpetual.

	Rearden Metal	Taggart Trans.
Before tax operating income	**655**	690
Interest on debt	**85**	105
Cost of capital (WACC)	**9-9.1 %**	8.625 %
Cost of equity capital	**15 %**	**13 %**
Cost of debt capital	**2.5 %**	**3 %**
Value of equity	**3800**	4500
Value of debt	3400	**3500**
Value of firm	7200	**8000**
Beta of equity	1.3	**1.1**
Beta of debt	0.05	**0.1**

Question 2.b – calculations for 2.a

Answers for Taggart Transcontinental:

Value of the Firm = ($690-$105+$105)/0.08625 = $8000

Debt = VF - Equity = $3500

Debt = $3500 = $105/rD ⇒ rD = 3%

Equity = $4500 = ($690-$105)/rE ⇒ rE = 13%

βE = (rE - rf)/(rMkt - rf) = (0.13 - 0.02)/(0.12 - 0.02) = 1.1

rU = rWACC = 0.13×$4500/$8000 + 0.03×$3500/$8000 = 8.625% (because tax is 0)

βU = (rU - rf)/(rMkt - rf) = (0.08625 - 0.02)/(0.12 - 0.02) = 0.6625

βD = βU - (βU - βE)×E/D = 0.6625 - (0.6625 - 1.1)×$4500/$3500 = 0.1

Answers for Rearden Metal:

Equity = VF - Debt = $3800

rE = rf - βE×(rMkt - rf) = 0.02 + 1.3×(0.12 - 0.02) = 15%

βU = βE×E/(E+D) + βD×D/(E+D) = 1.3×(3800/7200) + 0.05×(3400/7200) = 0.709722

rWACC = rU = rf - βU×(rMkt - rf) = 0.02 + 0.709722×(0.12 - 0.02) = 9.09722%

rD = rU + (rU - rE)×E/D = 0.0909722 + (0.0909722 - 0.15)×3800/3400 = 2.5%

Interest on debt = D×rD = $3400×0.025 = $85

Before tax operating income = E×rE + D×rD = $3800×0.15 + $85 = $655 (because tax is 0)

Question 2.c

Answer this question using the information given in 2.a.

Suppose that because of the large need for steel in building railroad infrastructure, Taggart Transcontinental and Rearden Metal decide to form into one large conglomerate. What is your estimate of the asset beta for this new conglomerate?

Taggart debt beta calculated from the table is 0.10.

βU = E/V βE + D/V βD = 4500/8000 × 1.1 + (8000-4500)/8000 × 0.10 = 0.6625

Rearden debt beta from the table is 0.05.

βU = E/V βE + D/V βD = 3800/7200 × 1.3 + (7200-3800)/7200 × 0.05 = 0.709722

Beta conglomerate = W_TT + W_RM

= 8000/(8000+7200) × (0.6625) + 7200/(8000+7200) × (0.709722) = 0.684868

Question 2.d

To form the conglomerate mentioned in (2.c) Taggart Transcontinental has issued new debt of the amount of $7200, to pay Rearden Metal's debt and buy the equity of the company at market value.

After the merger, however, the Taggart's executives realize that the company will not be able to pay back the total debt. Bankruptcy costs amount to $415. How much would the creditors of Taggart receive in the event of a bankruptcy? If the company would offer its creditors a fraction of its equity, how big would this fraction have to be for this deal to be accepted? Assume that the continuation value for the company in this case is the sum of all equity.

	Before Merger	After Merger
Taggart original debt	3500	3500
Taggart new debt	-	7200
Taggart equity value	4500	4500
Rearden original debt	3400	-
Rearden equity value	3800	3800

If bankrupt:

Total equity: 8300

(Minus) Bankruptcy cost: 415

Value to creditors: 7885

Which is the value that they would expect to receive.

If they take equity, then it would be:

value to creditors = equity participation × continuation value

7885 = e × 8300

e = 95%

Question 2.e

Assuming perfect capital markets Modigliani and Miller showed that whether firms are financed by equity and/or debt does not impact their value. Name 2 market imperfections that lead to the fact that the capital structure affects firm value and explain the impact on that value.

Adapted from Berk and DeMarzo, page 617 of the fourth edition.

The optimal capital structure depends on market imperfections, such as taxes, financial distress costs, agency costs, and asymmetric information.

Of all the different possible imperfections that drive capital structure, the most clear-cut, and possibly the most significant, is taxes. The interest tax shield allows firms to repay investors and avoid the corporate tax.

While firms should use leverage to shield their income from taxes, how much of their income should they shield? If leverage is too high, there is an increased risk that a firm may not be able to meet its debt obligations and will be forced to default. While the risk of default is not itself a problem, financial distress may lead to other consequences that reduce the value of the firm. Firms must, therefore, balance the tax benefits of debt against the costs of financial distress.

Agency costs and benefits of leverage are also important determinants of capital structure. Too much debt can motivate managers and equity holders to take excessive risks or under-invest in a firm. When free cash flows are high, too little leverage may encourage wasteful spending.

A firm must also consider the potential signaling and adverse selection consequences of its financing choice. Because bankruptcy is costly for managers, increasing leverage can signal managers' confidence in the firm's ability to meet its debt obligations.

Finally, it is important to recognize that because actively changing a firm's capital structure (for example, by selling or repurchasing shares or bonds) entails transactions costs, firms may be unlikely to change their capital structures unless they depart significantly from the optimal level.

Question 3.a

Consider the Pro Forma Balance Sheet for Ideko, 2008-2010, expressed in $1000's.

Year	2008	2009	2010
Balance Sheet ($ 000)			
Assets			
1 Cash and Cash Equivalents	9845	11,355	13,030
2 Accounts Receivable	19,689	22,709	26,059
3 Inventories	8854	10,240	11,784
4 **Total Current Assets**	38,388	44,304	50,872
5 Property, Plant, and Equipment	61,781	69,102	69,392
6 Goodwill	72,332	72,332	72,332
7 **Total Assets**	172,501	185,738	192,597
Liabilities			
8 Wages Payable	1941	2211	2570
9 Other Accounts Payable	5938	6900	7878
10 Total Current Liabilities	**7879**	**9110**	**10,448**
11 Debt	**115,000**	**120,000**	**120,000**
11 Total Liabilities	122,879	129,110	130,448

Calculate the amount of the increase in net working capital for Ideko in years 2009 and 2010.

Answer:

Year	2008	2009	2010
Working Capital ($ 000)			
Assets			
1 Cash and Cash Equivalents	9845	11,355	13,030
2 Accounts Receivable	19,689	22,709	26,059
3 Inventories	8854	10,240	11,784
4 Total Current Assets	38,388	44,304	50,872
Liabilities			
8 Wages Payable	1941	2211	2570
9 Other Accounts Payable	5938	6900	7878
10 Total Current Liabilities	7879	9110	10,448
Net Working Capital			
NWC **(4 - 10)**	30,509	35,194	40,425
Increase in NWC		4685	5231

Increase in NWC = NWC_t - NWC_{t-1}

Question 3.b

Using the values calculated in (3.a) and the following information (expressed in $1000's), calculate Ideko's Free Cash Flow to the Firm and Free Cash Flow to Equity for the same years.

Year	2009	2010
Income Statement($ 000)		
1 **Sales**	138,149	158,526
2 Cost of Goods Sold		
3 Raw Materials	(28,333)	(32,193)
4 Direct Labor Costs	(35,834)	(41,925)
5 **Gross Profit**	73,982	84,407
6 Sales and Marketing	(27,630)	(31,705)
7 Administrative	(17,959)	(20,608)
8 **EBITDA**	28,393	32,094
9 Depreciation	(7678)	(7710)
10 **EBIT**	20,715	24,383
11 Interest Expense (net)	(7820)	(8160)
12 **Pre-tax Income**	12,895	16,223
13 Income Tax	(4513)	(5678)
14 **Net Income**	**8382**	**10,545**

Year	2009	2010
Capital Expenditures	15,000	8000
Net Borrowing	5000	0

Answer:

Year	2009	2010
Net Income ($ 000)	8382	10,545
Plus: After Tax-Interest Expense	5083	5304
Unlevered Net Income	13,465	15,849
Plus: Depreciation	7678	7710
Less: Increases in NWC	(4685)	(5230)
Less: Capital Expenditures	(15,000)	(8000)
Free Cash Flow of Firm	1458	10,329
Plus: Net Borrowing	5000	0
Less: After-Tax Interest Expense	(5083)	(5304)
Free Cash Flow to Equity	1,375	5,025

After-Tax Interest Expense = Interest Expense × (1 - τc)

τc is 35% and can be computed from the income statement.

Question 3.c

The following are financial ratios for three companies which are comparable to Ideko:

Ratio	**Oakley, Inc.**	**Luxottica Group**	**Nike, Inc.**
P/E	24.8x	28x	18.2x
EV/Sales	2x	2.7x	1.5x
EV/EBITDA	11.6x	14.4x	9.3x

Based upon the average EV/EBITDA ratio of the comparable firms, and assuming that Ideko holds $6.5 million of cash in excess of its working capital needs, and has $4.5 million of debt in the year 2010, calculate Ideko's target market value of equity in 2010?

Answer:

Average EV/EBITDA = 11.77

EV = EV/EBITDA × EBITDA = 11.77 × $32.094 million = $377.75 million

EV = Equity + Debt - Cash in excess of NWC needs

Equity = EV - Debt + Cash in excess of NWC needs = $377.75 million - $4.5 million + $6.5 million = $379.75 million

Question 3.d

Assume that, given Ideko's future expected growth rate of 5%, the expected free cash flow for 2011 is $11 150 650.00.

If the company's WACC is 9%, calculate the continuation value (expressed in $1000's) in 2010.

V2010 = FCF2011 / WACC - g = 11,150,650 / (0.09 - 0.05) = $278,991,300

Question 4.a

The current price of Ocean Corporation stock is $6. In each six-month period this stock price can either go up by $2.50 or go down by $2. The stock pays no dividends. The one-year risk-free interest rate is 5%. Fill in Ocean stock price movements in a two-period binomial tree below.

Period 0	Period 1	Period 2
		S_{uu} = 11
	S_u = 8.5	
S = 6		S_{ud} = 6.5
	S_d = 4	
		S_{dd} = 2

Question 4.b

i) Calculate the payoffs for a currently <u>at the money one year call option</u> on the Ocean stock in the last period below.

ii) Sketch the payoff of a <u>one year call option</u> as a function of the strike price in the case that the price moved up during both six month periods.

i) The call is at the money, thus strike price K = 6. The payoffs at the end nodes are given by the formula: max(S – K, 0), in which S is the stock price. Hence, C_{uu} = max(11 – 6, 0) = 5, C_{ud} = max(6.5 – 6, 0) = 0.5 and C_{dd} = max(2 – 6, 0) = 0.

ii) The price of the stock after two up movements is equal to 11. Therefore, the answer is to sketch the function C_{uu} = max(11 – K, 0), with strike price K on the x-axis, call payoff C_{uu} on the y-axis. (I.e. a straight line through the points (0, 11) and (11, 0) and a zero value to the right of (11, 0).)

Question 4.c

Using the Binomial Model, calculate the price of a one-year European put option on the Ocean stock with a strike price of $7.

Given the annual risk free interest rate $r_f = 0.05$, the 6-monthly rate is given by $\hat{r}_f = (1 + r_f)^{\frac{1}{2}} - 1 \approx 0.0247$. Furthermore, the put payoffs at the final nodes are given by: P_{uu} = max(7 – 11, 0) = 0, P_{ud} = max(7 – 6.5, 0) = 0.5 and P_{dd} = max(7 – 2, 0) = 5.

The replicating portfolio of the option at node u is given by holding Δ_u of the stock and B_u of the bond.

$$\Delta_u = \frac{P_{uu} - P_{ud}}{S_{uu} - S_{ud}}, B_u = \frac{P_{ud} - S_{ud} \cdot \Delta_u}{1 + \hat{r}_f}$$

In the above P_i (S_i) denotes the value of the put (stock) at node i, and $\hat{r}_f$ is the previously calculated 6-month risk free rate. The (European) put price at node u is then given by

$$P_u = \Delta_u \cdot S_u + B_u$$

Plugging in the numbers for the up node u calculations:

$$\Delta_u = \frac{0 - 0.5}{11 - 6.5} = -\frac{1}{9}, \mathrm{B}_u = \frac{0.5 - 6.5 \cdot -\frac{1}{9}}{1 + 0.0247} \approx 1.1928$$

$$P_u \approx -\frac{1}{9} \cdot 8.5 + 1.1928 \approx 0.2484$$

Similarly, for the down node d:

$$\Delta_d = \frac{P_{ud} - P_{dd}}{S_{ud} - S_{dd}} = \frac{0.5 - 5}{6.5 - 2} = -1, B_d = \frac{P_{dd} - S_{dd} \cdot \Delta_d}{1 + \hat{r}_f} = \frac{5 - 2 \cdot -1}{1 + 0.0247} \approx 6.831$$

$$P_d = \Delta_d \cdot S_d + B_d \approx -1 \cdot 4 + 6.831 = 2.831$$

Finally, for the starting node:

$$\Delta = \frac{P_u - P_d}{S_u - S_d} \approx \frac{0.2484 - 2.831}{8.5 - 4} \approx -0.5739, B = \frac{P_d - S_d \cdot \Delta}{1 + \hat{r}_f} \approx \frac{2.831 - 4 \cdot -0.5739}{1 + 0.0247} \approx 5.003$$

$$P = \Delta \cdot S + B \approx -0.5739 \cdot 6 + 5.003 \approx 1.56$$

Therefore, the one-year European put is worth approximately $1.56.

Alternative approach: use the above approach to determine the call with strike K = 7, which is approx. 0.893. Then, use $P = C + PV(K) - S_0$ *to determine the value of the put.*

Question 4.d

Compute the risk neutral probability that the stock price will increase over the first six-month period. Assuming that investors are risk-averse, is the risk-neutral probability in the up state lower or larger than the actual probability in the upstate. Explain your answer.

$$S = \frac{S_u \cdot \rho + S_d(1 - \rho)}{1 + \hat{r}_f} \Leftrightarrow \rho = \frac{(1 + \hat{r}_f)S - S_d}{S_u - S_d} = \frac{1.0247 \cdot 6 - 4}{8.5 - 4} \approx 0.4773$$

Hence, the risk-neutral probability is approximately 0.477.

The risk-neutral probability in the up state is lower than the actual probability in the upstate. To ensure that all assets in the risk-neutral world have an expected return equal to the risk-free rate, relative to the true probabilities, the risk-neutral probabilities overweight the bad states and underweight the good states.

More detailed: In the real world investors are risk-averse. Thus, the expected return of a typical stock includes a positive risk premium to compensate investors of risk. IN the hypothetical risk-neutral world, investors do not require compensation for risk. So for the stock prices to be the same as in real world, investors must be more pessimistic. Thus, stocks that in reality have expected returns above the risk-free rate, when evaluated using these more pessimistic probabilities, have expected returns that equal the risk-free rate.

Question 5.a

You are a risk manager for Security First Trust Savings and Loan (SFTSL). SFTSL's balance sheet is as follows (in millions of dollars).

The duration of the auto loans is three years and the duration of the mortgages is eight years. Both cash reserves and checking and savings have zero duration. The CDs have a duration of two years and the long-term financing has a ten-year duration.

Assets		**Liabilities**	
Cash Reserves	100	Checking & Savings	200
Auto Loans	200	Certificates of Deposit	150
Mortgages	300	Long-Term Financing	200
Total Assets	600	|Total Liabilities	550
		Owner's Equity	50
		Total Liabilities and Equity	600

Calculate the duration of SFTSL's equity.

Answer: 14 years.

Explanation: D_{assets} = 100/600 × 0 (cash) + 200/600 × 3 (auto) + 300/600 × 8 (mortgage) = 5 years

$D_{liabilities}$ = 200/550 × 0 (checking) + 150/550 × 2 years (CDs) + 200/550 × 10 years (Long Term) = 4.181818 years

$D_{equity} = D_{assets} - D_{liabilities} = A/(A-L) \times D_A - L/(A-L) \times D_L$

= 600/(600-550) × (5 years) - 550/(600-550) × (4.181818 years) = 14.00 years

Question 5.b

Assume the data in (5.a) holds for an interest rate of 5%. Calculate the percentage change in SFTL equity if the interest rate falls to 4%.

Answer: 14% increase.

Explanation:

If interest rates fall by 1%, and given the duration calculated in (5.a), the value of SFTSL's equity will increase by about 14%.

Also accepted answers:

"The duration can also be interpreted as a measurement of the sensitivity to changes in interest rates. Hence, if the duration is equal to 14, it means that the value of the security/asset changes with 14 times the given change in the interest rate.

Thus, since the interest rate is decreasing by 1%, the SFTL equity increases by 14%."

"Modified duration = D/(1+r) = 14.02/1.05 = 13.35

The modified duration can be seen as the percentage decrease of the value per 1% increase in interest rate

percentage change in equity = - percentage change in interest rate*modified duration

Percentage change in equity = -(-1*13.35) = 13.35

The equity value will increase by 13.35% with an interest fall of 1%"

"If the interest falls to 4%, it means that the change of interest is -1%. We know the duration of assets and liabilities from 5a, which was

Duration of assets = 5

Duration of liabilities = 4.182

The duration can be used as a multiplier for change, meaning that for instance a 1% change in interest will lead to a -5% change in assets. Using this, we get the following new values:

$$Value_{assets} = Value_{assets,\,before} + Value_{assets,\,before} \cdot -5 \cdot change_{interest\ rate}$$
$$= 600\ +\ 600 \cdot (-5) \cdot (-1\%) = 630$$

$$Value_{liabilities} = Value_{liabilities,\,before} + Value_{liabilities,\,before} \cdot -4.182 \cdot change_{interest\ rate}$$
$$= 550 + 550 \cdot (-4.182)\,(-1\%) = 573$$

As equity = assets - liabilities, we have $Equity_{new} = 630 - 573 = 57$

The change in equity is then $\Delta Equity\ =\ \frac{57}{50} - 1 = 14\%$

The change in equity if interest falls to 4% will be 14%"

"To calculate the percentage change, I use duration approximation, where delta(equity) = equity(0)*delta(r)*(-D)

Change in equity = 50*(-1%)*(-14)=7

New equity = 50+7=57

Percentage change = 7/50=14%"